

Stream Control Transmission Protocol (SCTP)

Module 28

Duration: 26:43

Video: <https://www.youtube.com/watch?v=a277VGgcYi0>

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0:00 Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture

0:09 we will study about stream control transmission protocol that is sctp first of all please see the

0:17 contents which I shall cover after covering brief introduction about sctp and sctp Services I shall

0:25 take a features of sctp which will be followed by sctp packet format and last

0:33 I shall take up sctp Association so now let us start stream control transmission

0:41 protocol is a new reliable message oriented transport layer protocol figure shows the relationship of sctp to the

0:50 other protocols TCP and UDP in the Internet Protocol suit sctp lies between the application

0:58 layer and the n Network layer and serves as the intermediary between the application programs and the network

1:07 operations sctp however is mostly designed for internet applications that have recently been

1:15 introduced these new applications such as iua ISDN over IP

1:22 m2ua and M3 UA that is telephony signaling h248 that is Media India Gateway control

1:31 h.323 IP telephon N Sip IP telephony need a more sophisticated service that TCP

1:40 can provide sctp provides this enhanced performance and reliability we briefly

1:48 compare UDP TCP and sctp UDP is a message oriented protocol

1:57 process delivers a message to UDP which is encapsulated in a user datagram and

2:04 sent over the network UDP conserves the message boundaries each message is

2:10 independent from any other message this is a desirable feature when we are dealing with the applications such as IP

2:18 telephon and transmission of realtime data as we will see later in the text

2:25 however UDP is unreliable the sender cannot know the destination of messages

sent a message can be lost duplicated or

2:35 received out of order UDP also lacks some other features such as congestion control and flow control needed for a

2:44 friendly transport layer protocol TCP is a bite oriented protocol it is

2:52 receives a message or messages from a process stores them as a stream of bites and sends them in segment

3:01 3 minutes, 1 second there is no preservation of the message boundaries however TCP is a reliable protocol the duplicate segments are

3:09 detected the LW segments are recent and the bites are delivered to the end process in

3:15 order TCP also has congestion control and flow control mechanisms sctp combines the best

3:25 features of UDP and TCP sctp is a reliable message oriented protocol it preserves the message

3:34 boundaries and at the same time detects lost data duplicate data and out of order data it also has congestion

3:43 control and flow control mechanisms later we will see that sctp has other Innovative features

3:51 unavailable in UDP and TCP sctp is a message oriented reliable

3:59 protocol that combines the best features of UDP and TCP sctp

4:07 Services let us explain the services offered by sctp to the application layer processes process to process

4:15 communication sctp uses all whenn ports in the TCP space following table lists

4:23 some extra port numbers used by sctp protocol iua uses port number

4:32 90 protocol M2 UA uses port number

4:38 2904 protocol M3 UA users port number 2905 protocol

4:47 h248 uses port number 2949 protocol h.323 users port numbers

4:57 1718 1719 1 1720 and

5:03 11720 protocol sip users port number 5060 multiple

5:11 streams each connection between a TCP client and a TCP server involves one single stream the problem with this

5:20 approach is that a loss at any point in the Stream blocks the delivery of the rest of the

5:27 data this can be acceptable when we are transferring text it is not when we are sending realtime data such as audio or

5:36 video sctp allows multistream service in each connection which is called Association in sctp

5:46 terminology if one of these streams is blocked the other streams can still deliver their data the idea is similar to multiple

5:55 Lanes on a highway each lane can be used for a different type of traffic for example one lane can be used for a

6:03 regular traffic another for car pools if the traffic is blocked for regular Vehicles car pool Vehicles can still

6:11 reach their destinations multi homing a TCP connection involves one source and one

6:20 destination IP address this means that even if the sender or receiver is a multihomed host connected to more than

6:29 one physical address with multiple IP addresses only one of these IP addresses per end can be utilized during the

6:37 connection an sctp association on the other hand supports multihoming

6:44 service the sending and receiving host can Define multiple IP addresses in each in for an association in this fault

6:54 tolerant approach when one path fails another interface can be used for data delivery without

7:01 7 minutes, 1 second interruption this fault tolerant feature is very helpful when we are sending and receiving a realtime playl such as

7:10 Internet telephon figure shows the idea of multihoming in the figure the client is

7:17 connected to two local networks with two IP addresses the server is also connected to two networks with two IP

7:27 addresses the client and the server can make an association using four different pairs of IP addresses however note that

7:36 in the current implementations of sctp only one pair of IP addresses can be chosen for normal communication the

7:44 alternative is used if the main choice fails in other words at present sctp

7:52 does not allow load sharing between different parts satp Association allows multiple

7:59 IP addresses for each end full duplex communication like TCP sctp offers full

8:08 duplex service where data can flow in both directions at the same time each sctp then has a sending and receiving

8:17 buffer and packets are sent in both directions connection oriented service

8:24 like TCP sctp is a connection oriented protocol how however in sctp a connection is called an

8:33 association when a process at site a wants to send and receive data from another process at site B the following

8:42 occurs the two actps establish an association between each other data are exchanged in both

8:52 directions the association is terminated reliable service if sctp like

9:00 TCP is a reliable transport protocol it uses an acknowledgement mechanism to

9:07 check the safe and sound arrival of data we will discuss this feature further in the section on error

9:15 control sctp features let us discuss the general features of

9:22 sctp transmission sequence number TSN the unit of data in TCP is a bite

9:31 data transfer in TCP is controlled by numbering bytes using a sequence number on the other hand the unit of data in

9:40 sctp is a data chunk which may or may not have a one to one relationship with

9:47 the message coming from the process because of fragmentation data transfer in sctp is

9:55 controlled by numbering the data chunks SC TP uses a transmission sequence

10:01 10 minutes, 1 second number TSN to number the data chunks in other words the TSN in sctp plays the

10:09 analogous role as the sequence number in TCP tsn's are 32 bits long and randomly

10:18 initialized between 0 and $2^{32} - 1$ each data chunk must carry the

10:26 corresponding TSN in its header in satp a data chunk is numbered using a

10:34 transmission sequence number stream identifier si in TCP there is only one stream in

10:44 each Connection in sctp there may be several streams in each Association each

10:51 stream inp needs to be identified using a stream identifier SI each data chunk must get car the SI

11:00 in its header so that when it arrives at the destination it can be properly placed in its stream the SI is a 16 bit

11:09 number starting from zero to distinguish between different streams sctp users and

11:18 Si stream sequence number SSN when a data chunk arrives at the

11:25 destination satp it is delivered to the appr at stream and in the proper order

11:32 this means that in addition to an SI sctp defines each data Chunk in each

11:39 stream with a stream sequence number SSN to distinguish between different data chunks belonging to the same stream

11:48 satp uses ssns packets in TCP a segment carries data

11:56 and control information data data are carried as a collection of bytes control information is defined by six control

12:05 Flags in the header the design of sctp is totally different data are carried as

12:12 data chunks control information as control chunks several control chunks and data chunks can be packed together

12:20 in a packet a packet in sctp plays the same role as a segment in

12:27 TCP TCP has segments sctp has packets we will discuss the format of sctp packet

12:36 in the next section for the moment let us briefly list the differences between an actp

12:43 packet and a TCP segment the control information in TCP is part of the header the control

12:51 information in SCP is included in the control chunks there are several types of control chunks each is used for

13:00 different purpose the data in TCP segment treated as one entity an actp packet can carry

13:09 several data chunks each can belong to different stream the options section which can be

13:17 part of TCP segment does not exist in an sctp packet options in satp are handled

13:26 by defining new chunk types the mandatory part of TCP header is 20

13:32 bytes while the general header in sctp is only 12 bytes the sctp header is shorter due to the

13:41 following an sctp sequence number TSN belongs to each data chunk and hence

13:48 is located in the chunks header the acknowledgement number and window size are part of each control chunk there is

13:57 no need for a head length field known as HL in the TCP segment because there are

14:04 no options to make the length of the header variable the sctp header length

14:12 is fixed 12 bytes there is no need for an urgent pointer inp as we will see later the

14:20 check sum in TCP is 16 bits in sctb it is 32 bits the verification tag in sctp

14:29 is an association identifier which does not exist in TCP in TCP the combination

14:37 of Ip and Port addresses defines a connection in STP we may have

14:43 multihoming using different IP addresses a unique verification tag is needed to Define each

14:52 Association TCP includes one sequence number in the header which defines the number of First white in the data

14:59 section an actp packet can include several different data chunks TSN is and

15:09 ssns Define each data chunk some segments in TCP that carry control information need to consume one

15:18 sequence number control chunks in actp never use a TSN is or SSN number these

15:27 three identifiers below belong only to data chunks not to the whole packet in sctp control information and

15:36 data information are carried in separate chunks in STP we have data chunks

15:43 streams and packets an association may send many packets a packet may contain several chunks and chunks May belong to

15:51 different streams to make the definitions of these terms clear let us suppose that a process a needs to send

16:00 11 messages to process B and three streams the first four messages are in the first stream the second three

16:08 messages are in the second stream and the last four messages are in the third stream although a message if long can be

16:16 carried by several data chunks we assume that each message fits into one data chunk therefore we have 11 data chunks

16:25 in three streams the application process delivers 11 messages to sctp where each

16:32 message is ear marked for the appropriate stream although the process could deliver one message from the first

16:40 stream and then another from the second we assume that it delivers all messages belonging to the first stream first all

16:50 messages belonging to the second stream next and finally all messages belonging to the last stream

16:59 we also assume that the network allows only three data chunks per packet which means that we need four packets as shown

17:07 in the figure data chunks in stream zero are carried in the first and part of the

17:14 second packet those in stream one are carried in second and the third packet those in stream two are carried in the

17:23 third and fourth packet no that each data chunk needs

17:30 three identifiers TSN SI and SSN TSN is a cultivate number and used as we will see

17:40 later for flow control and error control s defines the stream to which the chunk belongs SSN defines the chunk order in a particular stream in our example SSN

17:53 starts from zero for each stream acknowledgement number TCP acknowledgement numbers are bite

18:02 oriented and refer to the sequence numbers sctp acknowledgement numbers are chunk oriented they refer to the

18:11 TSN a second difference between TCP and sctp acknowledgements is the control information recall that this information

18:20 is part of the segment header in TCP to acknowledge segments that carry only control information TCP uses a sequence

18:29 number and acknowledgement number for example a Sy YN segment needs to be acknowledged by an AC

18:39 segment in sctp however the control information is carried by the control chunks which do

18:46 not need a TSN these control chunks are acknowledged by another control chunk of

18:53 the appropriate type for example an init control chunk is acknowledged by an init

19:02 a chunk there is no need for a sequence number or an acknowledgement

19:09 number in actp acknowledgement numbers are used to acknowledge only data chunks

19:16 control chunks are acknowledged by other control chunks if necessary packet

format in this section

- 19:26 we show the format of a packet and different types of chunks most of the information presented in this section
- 19:33 will become clear later this section can be skipped in the first reading or used only as a reference an actp packet has a
- 19:42 mandatory General header having 12 bytes and a set of blocks called chunks having variable length there are two types of
- 19:51 chunks control chunks and data chunks a control chunk controls and maintains the association a data chunk carries user
- 20:01 20 minutes, 1 second data in an sctp packet control chunks come before data
- 20:08 chunks General header the general header that is Packet header defines the end
- 20:15 points of each Association to which the packet belongs guarantees that the packet belongs to a particular
- 20:23 Association and preserves the Integrity of the contents of packet including the head itself there are four fields in the
- 20:32 general header Source port address this is a 16bit field that defines the port number of the process
- 20:40 sending the packet destination port address this is a 16bit field that defines the port
- 20:48 number of the process receiving the packet verification tag this is a number
- 20:55 that matches a packet to an association this PR prevents a packet from a previous Association from being mistaken
- 21:02 as a packet in this Association it serves as an identifier for the association it is repeated in every
- 21:10 packet during the association there is a separate verification used for each Direction in the association
- 21:19 checksum this 32 bit contains a crc32 checkm note that the size of the checksum
- 21:26 is increased from 16 bits to 32 bits inp to allow the use of crc32
- 21:35 chum chunks control information or user data are carried in chunks chunks have a
- 21:43 common layout as shown in the figure the first three fields are common to all chunks the information field
- 21:50 depends on the type of Chunk the important point to remember that sctp requires the information section to be a
- 21:59 multiple of four bytes if not padding bytes are added at the end of the section chunks need to terminate on a
- 22:08 32bit boundary the description of the common fields are as follows type this 8bit
- 22:17 field can Define up to 256 types of chunks only a few have been defined so far the rest are reserved for future use
- 22:27 see the table for list of chunks and their descriptions flag this 8bit field defines

special

22:36 flags that a particular chunk may need each bit has a different meaning depending on the type of

22:44 Chunk length since the size of information section is dependent on the type of Chunk we need to define the

22:52 chunk boundaries the 16bit field defines the total size of the Chunk in bytes including the type flag and

22:59 length Fields sctp Association sctp like TCP is a

23:07 connection oriented protocol however a connection in sctp is called an association to emphasize

23:16 multihoming a connection in actp is called an association association

23:24 establishment Association establishment in SC TP requires a four-way handshake

23:31 in this procedure or process normally a client wants to establish an association with another process normally a server

23:39 using actp as the transport layer protocol similar to TCP the actp server

23:47 needs to be prepared to receive any association association establishment

23:54 however is initiated by the client sctp association establishment is as shown in the

24:02 figure the steps in a normal situation are as follows the client sends the

24:09 first packet which contains an init chunk the verification tag VT of this

24:16 packet defined in general header is zero because no verification tag has yet been defined for this direction client to

24:24 server the init tag includes an initiation tag to be used for packets from the other direction server to

24:32 client the chunk also defines the initial TSN for this direction and

24:38 advertises a value for RND the value of RND is normally advertised in a

24:46 s chunk it is done here because sctp allows the inclusion of data Chunk in

24:53 the third and fourth packets the server must be aware of the available client

24:59 buffer size note that no other chunks can be sent with the first packet the server sends the second

25:08 packet which contains an init acknowledgement chunk the verification tag is the value of the initial tag

25:16 field in the inet chunk this chunk initiates the tag to be used in the other direction defines the

25:25 initial TSN for data flow from server to client and sets the servers

25:31 rewind the value of rewind is defined to allow the client to send a data chunk with the third packet the init

25:40 acknowledgement also sends a cookie that defines the state of server at this moment we will discuss the use of the

25:48 cookies shortly the client sends the third packet which includes a cookie Eco chunk
25:57 this is a very simple chunk that EOS without change the cookie sent by the server
satp allows the included of data
26:06 chunks in this packet the server sends the fourth packet which includes the cookie
26:13 acknowledgement chunk that acknowledges the receipt of cookie e chunk sctp
26:20 allows the inclusion of data chunks with this packet so friends this was all about
26:28 stream control transmission protocol sctp hope the concepts explained in this
lecture were understandable and helpful
26:38 hope to see you in the next lecture till then goodbye thank you